OUTCOME-BASED EDUCATION (OBE) SYSTEM

Complete Application Setup and Startup Guide

Standard Operating Procedure for Developers, System Administrators & DevOps Engineers

Compliance: HEC Pakistan / Washington Accord / NCEAC Accreditation Criteria

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Project Metadata	System Specification
Application Architecture	Full-Stack React 19 SPA + Node Express Backend with Vite Dev Middlewares
Database Architecture	Supabase PostgreSQL with Row-Level Security (RLS) & Local In-Memory Fallback
Default Development Port	Port 3000 (Hardcoded runtime constraint in server.ts and vite.config.ts)
Language & Tools	TypeScript 5.8+, Vite 8+, Tailwind CSS v4, Tsx Runtime, Bun / NPM
Target Runtime Environment	Linux / Docker / Google Cloud Run Container Environment
Primary Verification Test Suite	tests/verify-obe-workflow.ts (74 Automated End-to-End Test Passes)

Table of Contents

1. Project Overview & Architectural Topology
2. Prerequisites, Software Compatibility & System Dependencies
3. Required Tools, Accounts & Secrets Configuration
4. Source Code Retrieval, Directory Layout & File Manifest
5. Clean Installation Commands & Dependency Lockfile Resolution
6. Native Dependencies & Vite 8 / React 19 Build Troubleshooting
7. Development Server Startup via Node / Express (Port 3000)
8. Database Setup: Supabase PostgreSQL Provisioning & Schemas
9. Environment Configuration: Safe .env.example Specification
10. Authentication Architecture & Password Hashing Security
11. Database Migrations Manifest: 001, 002 & 003 Schemas
12. Safe Database Migration Execution Protocol
13. Row-Level Security (RLS) & Multi-Role Permission Audit
14. First Administrator Bootstrap Creation Protocol
15. Local Verification & Computational Test Execution
16. Production Build & Deployment Commands
17. Comprehensive Error Diagnostics & Troubleshooting Matrix
18. Database Connectivity, CORS & Middleware Remediation
19. Automated Testing & Academic Logic Validation Protocol
20. Enterprise Security Policies & Credential Hygiene
21. Production Deployment Architecture & Container Infrastructure
22. Clean Development Reset & Safe Teardown Protocol
23. Post-Setup Verification Checklist & Acceptance Sign-off

1. Project Overview & Architectural Topology

The Pakistan University Outcome-Based Education (OBE) Management System is an enterprise educational computing platform designed to enforce curriculum alignment, granular student marks mapping to Bloom taxonomy levels, cohort learning outcome calculations, independent attendance examination eligibility rules, and Continuous Quality Improvement (CQI) closed-loop remediation.

The codebase uses a full-stack architecture where Express.js mounts Vite development server middleware during development and serves compiled production static assets in production. All authentication requests, institutional policy modifications, assessment grading, and attendance recording communicate through authenticated JSON REST API endpoints on port 3000.

2. Prerequisites & Software Versions

Software / Tool	Minimum Required Version	Recommended Version	Purpose
Node.js	v20.10.0 LTS	v22.x LTS	JavaScript / TypeScript V8 runtime engine
NPM / Bun	NPM v10.0+ / Bun 1.1+	NPM 10.8+ / Bun 1.2	Package management and lockfile enforcement
TypeScript	v5.8.0	v5.8.2	Type-checking, interfaces, and module compilation
Vite	v8.0.0	v8.3.0	Next-generation ES module frontend bundler
PostgreSQL	v15.0	v16.0 (Supabase Managed)	Relational database engine with RLS and PL/pgSQL

3. Required Tools, Accounts & Secrets

To operate this system in a cloud-connected configuration, obtain a Supabase project reference and retrieve your API credentials from Project Settings -> API. In local development or isolated air-gapped environments, the server automatically defaults to an in-memory database store seeded with synthetic engineering curriculum records.

4. Source Code Retrieval & Navigation

```
# Clone the repository
git clone <repository-url> obe-system
cd obe-system

# Verify project layout
ls -la
```

5. Installation & Dependency Resolution

Install all project dependencies declared in package.json. If dependency peer conflicts arise with React 19, use the strict resolution flag:

```
# Standard installation
npm install

# Or with Bun package manager (instant resolution)
bun install
```

7. Frontend Development Server Startup

The dev script starts the Express backend at server.ts using tsx, mounting Vite middlewares on port 3000:

```
# Run development environment
npm run dev

# Server output verification:
# [server.ts] Server running on http://localhost:3000
```

8. Database & Supabase Environment Setup

Configure the local .env file by copying the provided .env.example template. Never commit real secret keys into version control.

```
# Create .env from template
cp .env.example .env

# .env Configuration Keys:
PORT=3000
NODE_ENV=development
JWT_SECRET=super-secret-key-change-in-production-min-32-chars
VITE_SUPABASE_URL=https://your-project.supabase.co
VITE_SUPABASE_ANON_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
SUPABASE_SERVICE_ROLE_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
```

11. Database Migrations Manifest

The database schema is managed via incremental SQL scripts located in the supabase/migrations/ directory:

Migration File	Version	Primary Scope & Features Introduced
001_initial_schema.sql	v1.0.0	Initial OBE tables: departments, programs, courses, clos, plos, peos, assessments, marks, attendance, cqi_actions, users, and core RLS policies.
002_remove_framework_requirement.sql	v1.1.0	Removes hardcoded regulatory framework constraints from academic_policies to support adaptable university policies.
003_configurable_academic_policies_and_attendance_eligibility.sql	v2.0.0	Introduces hierarchical scoping columns (course_id, program_id, department_id, rule_purpose, default_action), stored function fn_get_effective_academic_policy, independent attendance calculator fn_evaluate_student_attendance_eligibility, and unified reporting view v_student_attendance_eligibility.

12. Safe Migration Execution Protocol

```
# Option A: Execute via Supabase CLI
supabase db push

# Option B: Direct psql execution
psql -h db.your-supabase.co -U postgres -d postgres -f supabase/migratio
ns/003_configurable_academic_policies_and_attendance_eligibility.sql
```

13. Row-Level Security & Role Access Audit

Row-Level Security (RLS) is enabled on all sensitive institutional tables. Students have SELECT-only access to their own enrollments, marks, and attendance. Teachers have write access to assessments and marks for course offerings assigned to them. Administrators have university-wide policy, curriculum, and user provisioning privileges.

14. First Administrator Bootstrap Protocol

When launching the system with a fresh, empty database (0 users registered), the login screen automatically detects this condition via /api/auth/status and presents the Initial Setup form. The administrator account is created securely with bcrypt password hashing via /api/auth/bootstrap-admin.

15. Local Verification & Computational Test Execution

Run the complete computational audit test suite before committing changes or deploying to staging:

```
# Execute the 74-pass verification suite
npm test

# Expected result:
# TEST EXECUTION SUMMARY: 74 PASSED, 0 FAILED
# All OBE verification workflows and calculation formulas PASSED with 100% precision.
```

16. Production Build & Deployment

```
# Build frontend assets
npm run build

# Start production server
npm run start
```

17. Error Diagnostics & Remediation Matrix

Observed Error / Symptom	Probable Root Cause	Standard Remediation Procedure
EADDRINUSE: port 3000	A zombie Node process is occupying port 3000	Run `fuser -k 3000/tcp` or `kill -9 $(lsof -t -i:3000)` and restart.
Missing JWT_SECRET	JWT_SECRET undefined in environment	Add a minimum 32-character secret to your .env file.
403 Forbidden on Marks Entry	Teacher attempting to grade unassigned offering	Verify course offering assignment in Sessions & Offerings tab.
Attainment combinedScore missing	No active academic policy configured	Navigate to Policy Settings tab as Admin and click "Add Policy" or "Set Default".
Attendance marked PENDING_EVALUATION	Course offering has 0 recorded attendance sessions	Record at least one session in Attendance Register before evaluating exam clearance.

23. Post-Setup Verification Checklist

Item #	Verification Milestone	Target Role	Pass Criteria
CHK-01	Authentication Status Check	Public / Guest	/api/auth/status returns valid JSON boolean.
CHK-02	Admin Account Provisioning	ADMIN	Master administrator registers and logs into Admin Dashboard.
CHK-03	Active Academic Policy	ADMIN	One university-wide policy is marked active with 80/20 weights.
CHK-04	Course Offering Setup	ADMIN	Course CS-301 Section A is mapped to Spring 2026 session.
CHK-05	Attendance Examination Debarment	TEACHER	Student with <75% attendance receives BARRED / INELIGIBLE status.
CHK-06	Attainment Computation	TEACHER / QEC	CLO attainment calculated accurately as achieved/eligible students.
CHK-07	CQI Loop Triggering	TEACHER	Deficient outcomes trigger NEEDS_CQI flag and log action in CQI module.
CHK-08	Official PDF Generation	ALL	SAR, Course File, and Transcript PDFs generate and download cleanly.